PROBLEM K5

Oscillating Planar Dipole

Original statement · K5.pdf

Velocity potential | Polar coordinates | Acceleration | Unsteady

OFFICIAL PROBLEM STATEMENT

For the planar fluid motion deriving from the velocity potential

$$\varphi = M\cos(\Omega t)\,\frac{\cos\theta}{r},$$

obtain:

1. Radial and azimuthal velocity components.
2. Expansion rate $\nabla \cdot \vec{v}$.
3. Vorticity.
4. Acceleration of the fluid particle initially located at $r = 1$ and $\theta = \pi/2$.
5. Initial value of the circulation around a circle of radius $R = 1/2$ centered at $(r = 1, \theta = \pi/2)$.
6. Streamlines.
7. Trajectories and particle paths.

Source: K5.pdf (page 1). Notes.pdf, Chapter 2: gradient, divergence and curl in orthogonal coordinates Eqs. (2.3), (2.5), (2.6); potential Eq. (2.34); material derivative and acceleration Eqs. (2.23)-(2.27); circulation and Stokes theorem Eqs. (2.29)-(2.33); trajectories Eq. (2.12); stream lines Eq. (2.21).

Worked solution (4-phase method)

PHASE 1

Hypotheses, Degrees of Freedom and Data

- Planar, unsteady, potential flow $\mathbf{v} = \nabla\varphi$ in the plane (r, θ), for $r > 0$. The potential is that of a dipole whose strength $M \cos \Omega t$ oscillates in time.
- Degrees of freedom: two spatial coordinates (r, θ) plus time; the only time scale is Ω^{-1}.
- Dimensions of the constants follow from φ (a velocity times a length, m^2/s): $M \cos \Omega t / r$ has the units of φ, so M has units m^3/s.
- The origin is a singular point and is excluded.

Symbol	Meaning	SI unit
M	dipole strength (per unit depth)	m^3/s
Ω	angular frequency	s^{-1}
r, θ	polar coordinates (the data $r = 1$, $R = 1/2$ are lengths in m)	m, rad
φ	velocity potential	m^2/s

PHASE 2

Coordinates, Frames and Changes of Variable

- Inertial frame at rest with the dipole; polar coordinates (r, θ) with scale factors $(1, r)$, $x = r\cos\theta$, $y = r\sin\theta$, and $\mathbf{e}_z = \mathbf{e}_r \wedge \mathbf{e}_\theta$.
- Gradient: $\nabla\varphi = \partial_r\varphi\, \mathbf{e}_r + r^{-1}\partial_\theta\varphi\, \mathbf{e}_\theta$; divergence: $\nabla \cdot \mathbf{v} = r^{-1}\left[\partial_r(r v_r) + \partial_\theta v_\theta\right]$; vorticity: $\omega_z = r^{-1}\left[\partial_r(r v_\theta) - \partial_\theta v_r\right]$ (Eqs. 2.3, 2.5, 2.6).
- Acceleration in curvilinear coordinates: the component form $a_i = Dv_i/Dt$ of Eq. (2.27) holds only in Cartesian coordinates. The coordinate-free form of Eq. (2.26), $\mathbf{a} = \partial_t\mathbf{v} + \nabla\left(|\mathbf{v}|^2/2\right) - \mathbf{v} \wedge (\nabla \wedge \mathbf{v})$, is used instead.
- For part 7 the material coordinates are the initial position (r_0, θ_0) at $t = 0$.

PHASE 3

Step-by-Step Derivation

3.1 Velocity components

Why this tool: since the motion derives from a potential, $\mathbf{v} = \nabla\varphi$ (Eq. 2.34), with the polar gradient above. Write $c(t) \equiv \cos\Omega t$.

$$v_r = \frac{\partial\varphi}{\partial r} = Mc\cos\theta\,\frac{\partial}{\partial r}\left(\frac{1}{r}\right) = \boxed{-\frac{Mc\cos\theta}{r^2},} \qquad v_\theta = \frac{1}{r}\frac{\partial\varphi}{\partial\theta} = \frac{1}{r}\cdot\frac{Mc}{r}\,(-\sin\theta) = \boxed{-\frac{Mc\sin\theta}{r^2}.}$$

3.2 Expansion rate

Why this tool: $\nabla\cdot\mathbf{v}$ is the rate of volume change per unit volume (Eqs. 2.62-2.63).

$$r\,v_r = -\frac{Mc\cos\theta}{r} \;\Rightarrow\; \frac{\partial(rv_r)}{\partial r} = \frac{Mc\cos\theta}{r^2}, \qquad \frac{\partial v_\theta}{\partial\theta} = -\frac{Mc\cos\theta}{r^2},$$

$$\nabla\cdot\mathbf{v} = \frac{1}{r}\left[\frac{Mc\cos\theta}{r^2} - \frac{Mc\cos\theta}{r^2}\right] = \boxed{0.}$$

The flow is incompressible for all t (consistent with $\nabla^2\varphi = 0$).

3.3 Vorticity

Why this tool: $\omega_z = r^{-1}\left[\partial_r(rv_\theta) - \partial_\theta v_r\right]$ is twice the angular velocity of the fluid element (Eq. 2.52).

$$r\,v_\theta = -\frac{Mc\sin\theta}{r} \;\Rightarrow\; \frac{\partial(rv_\theta)}{\partial r} = \frac{Mc\sin\theta}{r^2}, \qquad \frac{\partial v_r}{\partial\theta} = \frac{Mc\sin\theta}{r^2},$$

$$\omega_z = \frac{1}{r}\left[\frac{Mc\sin\theta}{r^2} - \frac{Mc\sin\theta}{r^2}\right] = \boxed{0.}$$

The motion is irrotational, as it must be for $\mathbf{v} = \nabla\varphi$ ($\nabla \wedge \nabla\varphi \equiv 0$).

3.4 Acceleration of the particle initially at $r = 1, \theta = \pi/2$

Why this tool: the acceleration is the material derivative of the velocity (Eq. 2.25). Because $\boldsymbol{\omega} = 0$, Eq. (2.26) reduces to $\mathbf{a} = \partial_t \mathbf{v} + \nabla(|\mathbf{v}|^2/2)$, which is valid in polar coordinates.

Local term, using $\partial_t c = -\Omega\sin\Omega t$:

$$\frac{\partial v_r}{\partial t} = \frac{M\Omega\sin\Omega t\cos\theta}{r^2}, \qquad \frac{\partial v_\theta}{\partial t} = \frac{M\Omega\sin\Omega t\sin\theta}{r^2}.$$

Convective term: $|\mathbf{v}|^2 = v_r^2 + v_\theta^2 = \dfrac{M^2c^2\left(\cos^2\theta + \sin^2\theta\right)}{r^4} = \dfrac{M^2c^2}{r^4}$, hence

$$\frac{|\mathbf{v}|^2}{2} = \frac{M^2c^2}{2r^4}, \qquad \frac{\partial}{\partial r}\left(\frac{|\mathbf{v}|^2}{2}\right) = -\frac{2M^2c^2}{r^5}, \qquad \frac{1}{r}\frac{\partial}{\partial\theta}\left(\frac{|\mathbf{v}|^2}{2}\right) = 0.$$

Field acceleration:

$$a_r = \frac{M\Omega\sin\Omega t\cos\theta}{r^2} - \frac{2M^2\cos^2\Omega t}{r^5}, \qquad a_\theta = \frac{M\Omega\sin\Omega t\sin\theta}{r^2}.$$

Evaluate at $r = 1, \theta = \pi/2$ ($\cos\theta = 0, \sin\theta = 1$):

$$\boxed{\mathbf{a}(1, \tfrac{\pi}{2}, t) = -2M^2\cos^2(\Omega t)\,\mathbf{e}_r + M\Omega\sin(\Omega t)\,\mathbf{e}_\theta.}$$

This is the acceleration of whichever particle occupies the point $(1, \pi/2)$ at time t. For the particle that is at that point at the initial instant, set $t = 0$: $\mathbf{a} = -2M^2\mathbf{e}_r$ (the local term vanishes because $\sin 0 = 0$). For $t > 0$ that particle has moved away from the point, so its

acceleration is the field value at its new position (part 7).

3.5 Initial circulation around the circle of radius $1/2$ centred at $(r = 1, \theta = \pi/2)$

Why this tool: $\Gamma = \oint \mathbf{v} \cdot d\mathbf{l}$ (Eq. 2.29) equals the flux of the vorticity through any surface bounded by the curve that lies in the fluid (Eq. 2.32). The centre is at Cartesian $(0, 1)$ and the circle has radius $1/2$, so every point of the disc is at a distance of at least $1/2$ from the origin: the disc lies entirely inside the domain, where $\omega_z = 0$.

$$\Gamma = \iint_S \omega_z \, d\sigma = \iint_S 0 \, d\sigma = \boxed{0}$$

at $t = 0$ and, since $\omega_z = 0$ for all t, at every instant. In fact φ is single-valued, so $\oint \nabla\varphi \cdot d\mathbf{l} = \varphi(\text{end}) - \varphi(\text{start}) = 0$ for every closed curve, including those that enclose the origin.

3.6 Streamlines

Why this tool: streamlines are tangent to $\mathbf{v}$ at a frozen instant (Eq. 2.21): $dr/v_r = r\,d\theta/v_\theta$. At an instant with $\cos\Omega t \neq 0$ the common factor $-Mc/r^2$ cancels:

$$\frac{dr}{\cos\theta} = \frac{r\,d\theta}{\sin\theta} \implies \frac{dr}{r} = \frac{\cos\theta}{\sin\theta}\,d\theta \implies \ln\frac{r}{r_0} = \ln\left|\frac{\sin\theta}{\sin\theta_0}\right|,$$

$$\boxed{r = \frac{r_0}{\sin\theta_0}\sin\theta = C\sin\theta.}$$

In Cartesian variables $r^2 = Cr\sin\theta$ gives $x^2 + y^2 = Cy$: circles of diameter C centred at $(0, C/2)$ and tangent to the x-axis at the origin. The shape does not depend on t. (At the isolated instants with $\cos\Omega t = 0$ the whole velocity field vanishes.) Equivalent form: the stream function $\psi = -Mc\sin\theta/r$ (with $v_r = r^{-1}\partial_\theta\psi$, $v_\theta = -\partial_r\psi$) has the level curves $\sin\theta/r = \text{const}$.

3.7 Trajectories and particle paths

Why this tool: trajectories solve $d\mathbf{x}/dt = \mathbf{v}$ (Eq. 2.12), here

$$\frac{dr}{dt} = -\frac{Mc\cos\theta}{r^2}, \qquad r\frac{d\theta}{dt} = -\frac{Mc\sin\theta}{r^2}, \qquad r(0) = r_0,\ \theta(0) = \theta_0.$$

Dividing the first equation by the second eliminates both t and the factor $c(t)$:

$$\frac{dr}{r\,d\theta} = \frac{\cos\theta}{\sin\theta} \implies \boxed{r = r_0\frac{\sin\theta}{\sin\theta_0}}$$

which is the path line. It coincides with the streamlines of 3.6 because the direction of $\mathbf{v}$ does not depend on time (only its magnitude does).

Time law. Substitute the path into the second equation, $r^3 = r_0^3\sin^3\theta/\sin^3\theta_0$:

$$\frac{d\theta}{dt} = -\frac{Mc\sin\theta}{r^3} = -\frac{M\sin^3\theta_0}{r_0^3}\,\frac{\cos\Omega t}{\sin^2\theta} \implies \sin^2\theta\,d\theta = -\frac{M\sin^3\theta_0}{r_0^3}\cos\Omega t\,dt.$$

Integrate with Barrow's rule between $(\theta_0, 0)$ and (θ, t). For the left side use $\sin^2\theta' = (1-\cos 2\theta')/2$, whose primitive is $\theta'/2 - \sin 2\theta'/4$; for the right side, $\int_0^t \cos\Omega t'\,dt' = \sin(\Omega t)/\Omega$:

$$\boxed{\frac{\theta-\theta_0}{2} - \frac{\sin 2\theta - \sin 2\theta_0}{4} = -\frac{M\sin^3\theta_0}{r_0^3\,\Omega}\sin\Omega t.}$$

Together with $r = r_0\sin\theta/\sin\theta_0$ this defines $\mathbf{x}(t)$ implicitly. The right-hand side vanishes whenever $\sin\Omega t = 0$, so the particle returns to its starting point every half period π/Ω: it oscillates back and forth along its circular path. (Particles on the axis $\theta_0 = 0$ or π stay on it, with $r^3 = r_0^3 \mp 3(M/\Omega)\sin\Omega t$.)

PHASE 4

Interpretation, Limits and Dimensional Check

- The flow is the oscillating field of a point dipole: incompressible, irrotational, with streamlines that are circles through the singular point and a velocity that decays as r^{-2}.
- Steady limit $\Omega \to 0$ (with $c \to 1$): the local term vanishes and the acceleration is purely convective, $a_r = -2M^2/r^5$ and $a_\theta = 0$, equal to $\nabla(|\mathbf{v}|^2/2)$; the particle then drifts monotonically along the circle.
- Consistency of the answers: part 5 ($\Gamma = 0$) follows from part 3 ($\omega_z = 0$); part 7 reproduces part 6 geometrically.
- Dimensions (SI): $[v] = [M/r^2] = (\mathrm{m}^3/\mathrm{s})/\mathrm{m}^2 = \mathrm{m/s}$; $[a] = [M^2/r^5] = (\mathrm{m}^6/\mathrm{s}^2)/\mathrm{m}^5 = \mathrm{m/s}^2$ and $[M\Omega/r^2] = (\mathrm{m}^3/\mathrm{s})(1/\mathrm{s})/\mathrm{m}^2 = \mathrm{m/s}^2$; the combination $M/(r_0^3\Omega)$ in 3.7 is $(\mathrm{m}^3/\mathrm{s})/(\mathrm{m}^3/\mathrm{s}) = 1$, dimensionless as required for an angle.
- Numerical cross-check: symbolic differentiation of φ gives $\nabla \cdot \mathbf{v} = 0$, $\omega_z = 0$ and, from the full material derivative in polar components (including the $-v_\theta^2/r$ and $v_r v_\theta/r$ terms), the same a_r, a_θ as above, in particular $\mathbf{a}(1, \pi/2, t) = -2M^2 \cos^2 \Omega t\, \mathbf{e}_r + M\Omega \sin \Omega t\, \mathbf{e}_\theta$. RK4 integration of the particle ODE with $M = 0.8$ m^3/s, $\Omega = 1.7$ s^{-1}, $r_0 = 1$ m, $\theta_0 = 0.45\pi$ up to $t = 1.3$ s gives $r = 0.846496$ m, equal to $r_0 \sin\theta / \sin\theta_0$ at the integrated angle, and the two sides of the boxed time law, -0.363902 and -0.363902.

> **READING OF THE OFFICIAL HANDWRITTEN SOLUTION**
>
> K5.pdf evaluates the acceleration at the field point $(1, \pi/2)$ for general t, obtaining the vector given in 3.4; the value for the particle that is there at $t = 0$ is its $t = 0$ value. The handwritten trajectory law agrees with the boxed relation in 3.7 and notes that streamlines and path lines coincide. No numerical discrepancy was found.